

VISION-AWARE DS-MODULATION OBSTACLE AVOIDANCE

Research Project Final Report

Perception-aware obstacle avoidance under realistic computer-vision degradation

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Abstract

Dynamical System (DS)-based obstacle avoidance provides a reactive framework for generating collision-free robot motion, but its behavior depends on the quality of the obstacle information supplied to the controller. In practical robotic systems, obstacle positions are obtained through computer-vision perception and can therefore be affected by measurement noise, blur, limited frame rate, latency, and occlusion. This project investigates how these perception imperfections affect DS-modulation-based obstacle avoidance and evaluates whether Kalman filtering can recover part of the resulting performance degradation.

A vision-based obstacle perception pipeline was developed and integrated with a Kalman filter and a DS-modulation controller in a 2D simulated robotic environment. Controlled experiments were performed under different levels of pixel noise, blur, frame rate, latency, and occlusion. Localization RMSE, minimum clearance, convergence time, path length, collision rate, and goal-reaching rate were used to evaluate the perception-control pipeline. The complete pipeline was subsequently integrated with ROS 2 and TurtleBot3 in Gazebo as an end-to-end demonstration.

The results show that increasing pixel noise increases localization error, while low frame rates and increased latency degrade temporal and control behavior. Blur had comparatively limited influence over the tested range. Kalman filtering consistently reduced localization RMSE under pixel noise and across the tested latency conditions, but it does not explicitly compensate for delay. Its benefit depended on perception update rate and was limited under systematic errors caused by occlusion. Overall, the study demonstrates that perception quality is an important factor in DS-modulation obstacle avoidance and that filtering stochastic measurement noise does not fully compensate for systematic perception errors.

1. Introduction

Reactive obstacle avoidance allows a robot to modify its motion online in response to obstacles without requiring a complete precomputed trajectory. Dynamical-system-based approaches are useful because a desired motion field can be continuously modified near obstacles while retaining attraction toward a goal. The obstacle-avoidance formulation used here follows the DS modulation framework introduced by Khansari-Zadeh and Billard and later developments in reactive dynamical-system obstacle avoidance.

A key practical assumption in many controlled demonstrations is that obstacle information is accurate. A camera-based robot instead receives an estimate generated by a perception pipeline. Errors in that estimate can change obstacle distance and the local obstacle normal used by the modulation law, and therefore can change the resulting robot motion.

This project connects perception quality with control behavior by introducing controlled perception degradation and measuring how the complete perception-to-control pipeline responds. A Kalman filter is then used to determine whether temporal state estimation can recover part of the performance lost due to noisy measurements.

1.1 Research Question

How does DS-modulation-based obstacle avoidance degrade when obstacle position estimates come from noisy, realistic computer-vision perception—such as blur, occlusion, latency, and low frame rate—instead of clean ground-truth data, and can a Kalman filter recover performance toward the clean baseline?

1.2 Objectives

- Develop a 2D robot simulation for goal-directed motion and obstacle avoidance.
- Build a camera-based obstacle perception pipeline with image-to-ground geometric conversion.
- Introduce controlled pixel noise, blur, frame-rate reduction, latency, and occlusion.
- Estimate obstacle position using a Kalman filter and compare filtered and unfiltered perception.
- Measure how perception errors propagate into DS-modulation obstacle avoidance.
- Evaluate localization and control behavior over repeated runs.
- Integrate the pipeline with ROS 2 and TurtleBot3 in Gazebo as a demonstration.

2. Background and Related Work

2.1 Dynamical-System Obstacle Avoidance

In a dynamical-system formulation, the desired robot motion is represented by a vector field that attracts the robot toward a goal. Obstacle avoidance is introduced by modifying this vector field near an obstacle. The modulation approach changes the velocity components along obstacle-normal and tangent directions, redirecting motion around the obstacle while allowing the nominal goal-directed dynamics to dominate away from the obstacle.

Later work by Huber, Billard, Slotine and collaborators extends reactive dynamical-system approaches to sparse or asynchronous sensing, nonconvex obstacles, and more complex environments. The focus of this project is complementary: it studies how errors in the perception supplied to the modulation controller affect the resulting behavior.

2.2 Vision and State Estimation

The perception pipeline identifies the obstacle in an image, extracts an image-space location, and converts it to a ground-plane position. A constant-velocity Kalman filter then predicts and corrects the planar obstacle state. This provides a principled comparison between raw perception and temporally filtered perception.

3. System Architecture

The project is organized as a layered perception-to-control pipeline:

Camera → OpenCV Detection → Pixel-to-Ground Projection → Kalman Filter → DS Modulation → Velocity Command → Robot

Layer	Function	Output
Perception	Detect obstacle in camera image	Image-space location
Geometry	Convert image coordinates to ground-plane position	Relative obstacle position
State estimation	Predict/update obstacle state	Filtered obstacle position
Control	Modify goal-directed DS near obstacle	Desired planar velocity
Robot interface	Send motion command	cmd_vel / TwistStamped
Evaluation	Measure perception/control behavior	RMSE, clearance, time, path, collision, goal rate

4. Methodology

4.1 Robot and Environment

The main research experiments used a 2D simulated robot moving toward a fixed goal in the presence of a single static convex obstacle. The robot was modeled in planar coordinates with a collision radius of 0.2 world units. The simulator kept the true obstacle state separate from the perceived obstacle supplied to the controller. Configurations in which the robot, obstacle center, and goal were exactly collinear were excluded because the current DS implementation has a known ambiguity in selecting an avoidance direction.

4.2 Goal-Directed Dynamics and DS Modulation

The nominal controller generates a velocity pointing from the robot toward the goal, with a proportional gain and speed limit. Near an obstacle, the desired velocity is transformed in a local basis formed by the obstacle normal and tangent directions. The modulation reduces the component directed toward the obstacle and changes the tangential component so that the resulting flow can pass around it. The influence decreases with distance.

The implementation uses an effective circular obstacle representation, includes robot-radius clearance, and retains a small tangential bias to avoid a zero tangential component in geometrically ambiguous cases.

4.3 Computer-Vision Perception

The perception pipeline uses OpenCV to segment the visible obstacle and extract its image-space location. The detected image point is converted to a ground-plane position. The research experiments use a simplified synthetic camera model with a linear image-to-world scale of 0.02 world units per pixel. This controlled mapping is used for the systematic degradation experiments. The ROS 2/Gazebo demonstration uses a separate camera-geometry projection based on the simulated camera parameters.

4.4 Pixel-to-Ground Projection

For the ROS 2 camera demonstration, the projection uses camera parameters taken from the simulated 320×240 camera: $f_x = 277$, $f_y = 277$, $c_x = 160$, and $c_y = 120$. The image point is back-projected as a camera ray and intersected with the robot ground plane. The resulting relative position is then transformed into the

world frame using the robot's planar pose. These parameters belong to the ROS demonstration and are not the pixel-to-world mapping used in the main synthetic experiments.

$$x_n = (u - c_x)/f_x \quad ; \quad y_n = (v - c_y)/f_y$$

$$p_{world} = p_{robot} + R(\psi) p_{relative}$$

4.5 Kalman Filter

The state contains planar obstacle position and velocity. A constant-velocity transition model predicts the state between measurements, while the image-derived planar position provides the measurement update. This standard Kalman filtering formulation follows the state-estimation framework introduced by Kalman [8]. The filter therefore primarily addresses stochastic measurement noise and provides a smoother obstacle estimate.

4.6 Perception Degradation

Degradation	Purpose	Levels
Pixel noise	Random image-space localization uncertainty	0, 2, 4, 6, 8, 10 px
Blur	Reduced image sharpness	0, 3, 5, 7, 9 px
Frame rate	Sparse perception updates	2, 4, 5, 10, 20 FPS
Latency	Delayed obstacle measurements	0, 50, 100, 150, 200 ms
Occlusion	Partial/missing visual information	0, 10, 20, 30, 40, 50%
Noise × occlusion	Interaction of stochastic/systematic error	0/5/10 px × 0/25/50%

4.7 Evaluation Metrics and Statistics

Metric	Meaning
Localization RMSE	Distance between estimated and true obstacle position.
Minimum clearance	Smallest robot-obstacle separation.
Convergence time	Time required to reach the goal tolerance.
Path length	Accumulated robot trajectory length.
Collision rate	Fraction of runs ending in collision.
Goal-reaching rate	Fraction of runs reaching the goal condition.

The main sweeps used repeated runs. Paired comparisons used paired t-tests where appropriate, with Holm correction for multiple comparisons within each experiment and $\alpha = 0.05$. The occlusion sweep was evaluated descriptively because repeated runs produced zero-variance paired differences for the reported metrics, making conventional paired tests degenerate.

5. Results

5.1 Pixel Noise

Pixel noise produced a clear increase in localization error. Unfiltered RMSE increased from 0.0548 at 2 px to 0.2755 at 10 px, while the Kalman-filtered RMSE increased more slowly from 0.0367 to 0.1766. After Holm correction, the Kalman filter significantly reduced localization RMSE at all non-zero noise levels ($p < 0.001$). A significant difference in minimum clearance occurred only at 10 px, and no significant differences were observed in convergence time or path length.

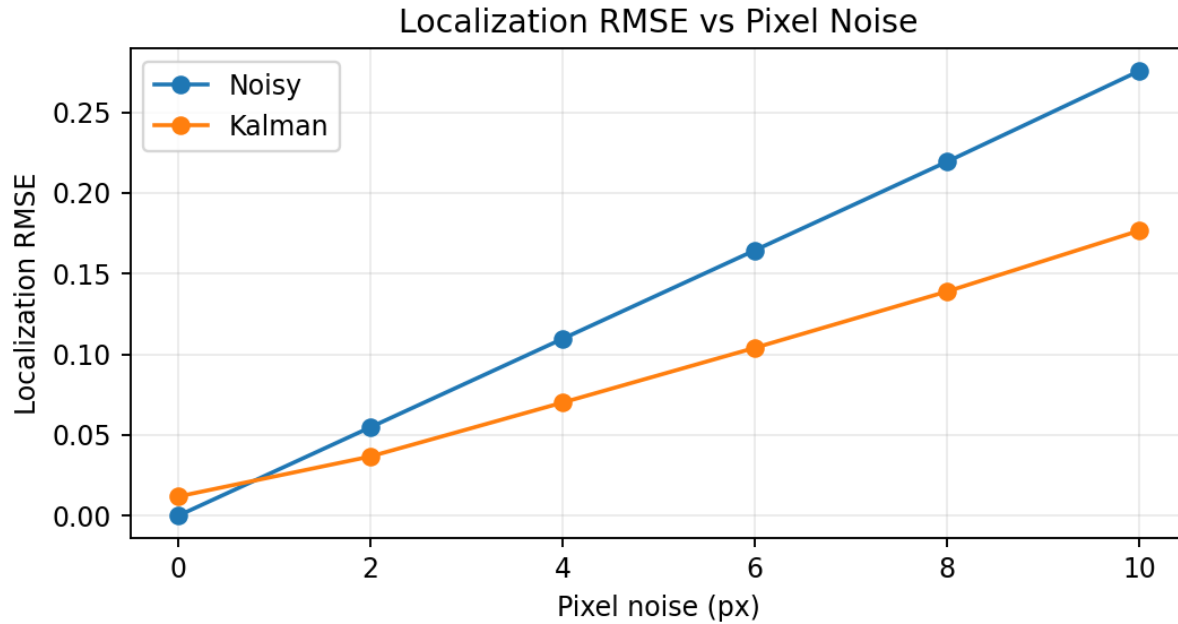


Figure 1. Localization RMSE vs pixel noise.

Noise	Noisy RMSE	Kalman RMSE	Noisy clearance	Kalman clearance	Collision N	Collision K
0	0.0000	0.0119	0.0933	0.0929	0%	0%
2	0.0548	0.0367	0.0913	0.0918	0%	0%
4	0.1096	0.0701	0.0872	0.0885	0%	0%
6	0.1644	0.1040	0.0806	0.0844	0%	0%
8	0.2193	0.1389	0.0714	0.0795	2%	2%
10	0.2755	0.1766	0.0609	0.0745	10%	8%

5.2 Blur

The tested blur levels had comparatively little effect on the unfiltered RMSE, which stayed near 0.137, while Kalman RMSE stayed near 0.087. The Kalman filter significantly reduced localization RMSE at all tested blur levels, but no statistically significant differences were found in clearance, convergence time, or path length. Blur levels 3–5 px and 7–9 px produced identical aggregate results; this is likely because the blur-and-threshold processing produced effectively identical binary masks in those conditions.

Blur	Noisy RMSE	Kalman RMSE	Noisy clearance	Kalman clearance	Noisy time	Kalman time
0	0.1369	0.0871	0.0842	0.0866	0.9625	0.9610
3	0.1370	0.0869	0.0930	0.0952	0.9740	0.9720
5	0.1370	0.0869	0.0930	0.0952	0.9740	0.9720
7	0.1370	0.0869	0.0966	0.0987	0.9785	0.9755
9	0.1370	0.0869	0.0966	0.0987	0.9785	0.9755

5.3 Frame Rate

Frame rate strongly affected the usefulness of filtering. At 2 and 4 FPS, Kalman RMSE was significantly higher than the noisy condition. At 5 FPS the difference was not statistically significant, while at 10 and 20 FPS the filter significantly reduced RMSE. In the tested setup, collision rates were 32%, 5%, 4%, 1%, and 0% at 2, 4, 5, 10, and 20 FPS respectively, with the same collision rates for the filtered and unfiltered conditions.

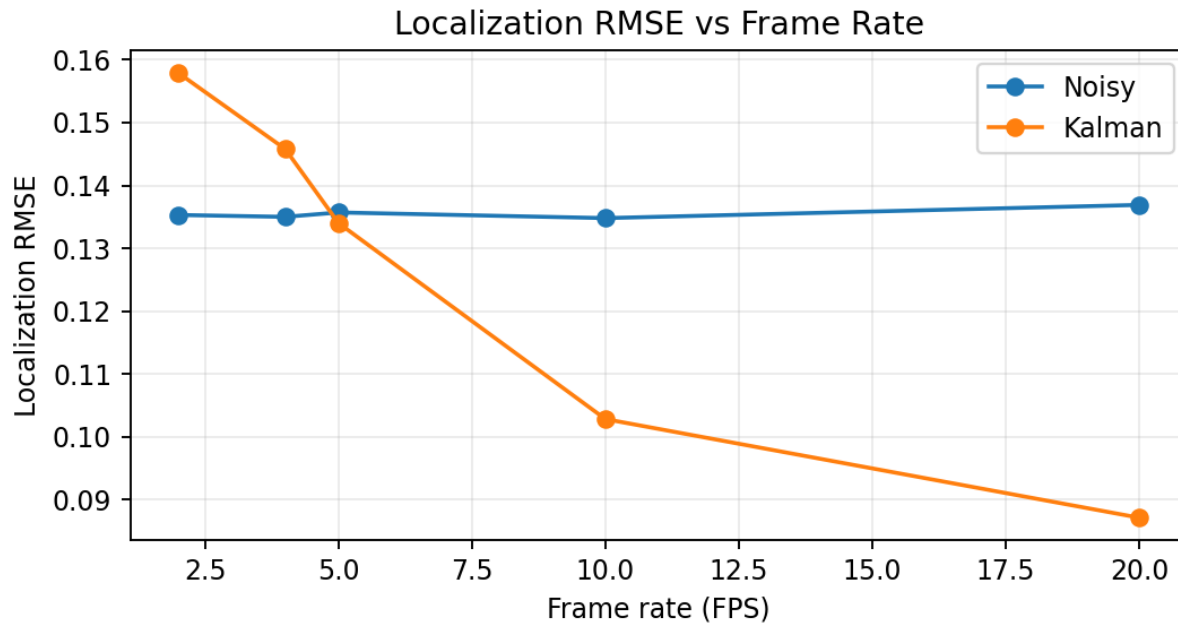


Figure 2. Localization RMSE vs frame rate.

FPS	Noisy RMSE	Kalman RMSE	Collision	Goal rate
2	0.1353	0.1579	32%	68%
4	0.1350	0.1458	5%	95%
5	0.1357	0.1340	4%	96%
10	0.1348	0.1028	1%	99%
20	0.1369	0.0871	0%	100%

5.4 Latency

Increasing latency reduced minimum clearance and increased convergence time. At 200 ms, minimum clearance was 0.0592 without filtering and 0.0737 with Kalman filtering. Collision rate increased from 0% at 0–50 ms to 9% noisy and 7% Kalman at 200 ms. The Kalman filter significantly reduced RMSE at every tested latency level ($p < 0.001$), with significant clearance differences at 150 and 200 ms. Filtering did not remove the degradation caused by delayed information. Importantly, the obstacle was static in these experiments, so the delayed estimate was compared with an unchanged true obstacle position; this makes localization RMSE largely insensitive to latency. The latency effect appears primarily in control, because the robot continues moving while the controller acts on delayed perception.

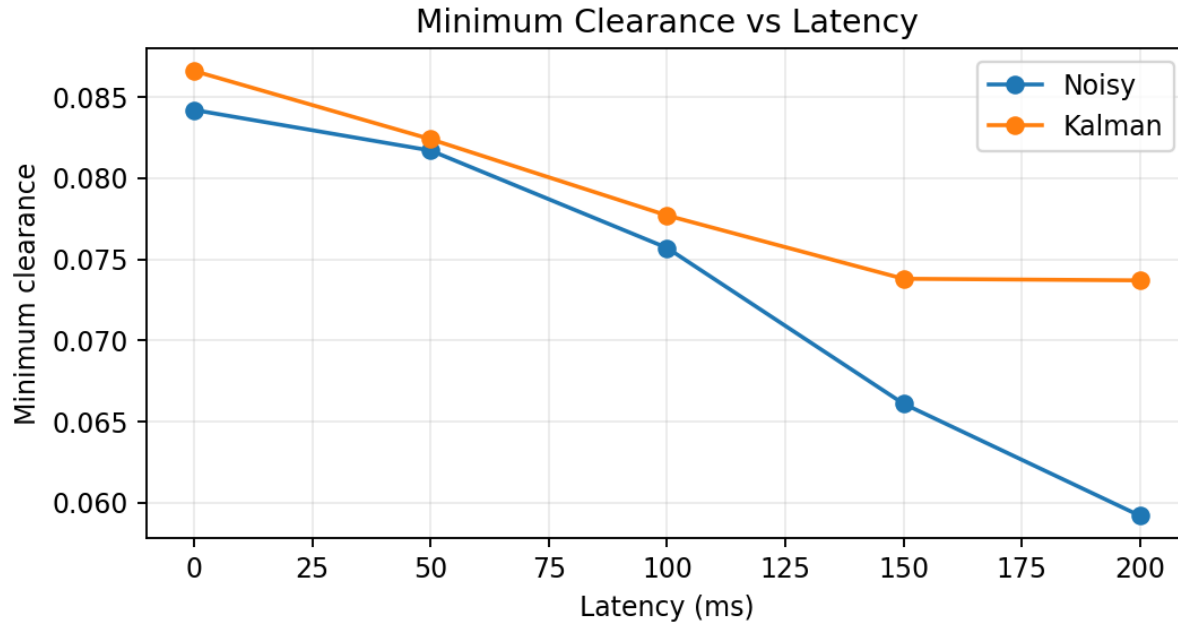


Figure 3. Minimum clearance vs latency.

Latency	Noisy RMSE	Kalman RMSE	Noisy clearance	Kalman clearance	Collision N	Collision K
0	0.1369	0.0871	0.0842	0.0866	0%	0%
50	0.1368	0.0866	0.0817	0.0824	0%	0%
100	0.1374	0.0867	0.0757	0.0777	1%	1%
150	0.1382	0.0870	0.0661	0.0738	6%	3%
200	0.1378	0.0875	0.0592	0.0737	9%	7%

5.5 Occlusion

Occlusion introduced systematic errors rather than merely random measurement noise. In the tested fixed geometry, the Kalman-filtered condition resulted in collisions at 10–40% occlusion while the noisy condition reached the goal. At 50% occlusion, the detected-position bias changed and both conditions reached the goal without collision. These repeated occlusion outcomes were deterministic in the tested setup, so the reported 100% and 0% collision rates should be interpreted as repeatability within one fixed obstacle geometry and occlusion model, not as general collision probabilities across environments.

Mechanism evidence: Additional diagnostics examined the obstacle normals used by the controller. At 25% occlusion, the true normal was approximately $(-0.956, -0.294)$, while the estimated normals across seeds 2–4 were $(-0.988, -0.152)$, $(-0.986, -0.164)$, and $(-0.974, -0.226)$; these cases resulted in collision. At 50% occlusion, the centroid bias changed direction and the estimated normals were different, yet the tested trajectories did not collide. Across seeds 2–4, the detected x-coordinate was consistently greater than the true x-coordinate at 10–40% occlusion and consistently smaller at 50%. The representative normal comparisons in Table 1 were obtained in the Kalman condition with 5 px pixel noise.

Open question: the diagnostics do not isolate why the noisy and Kalman conditions diverged when their aggregate RMSE values were nearly identical (for example, 0.4346 versus 0.4350 at 10% occlusion). A smoothing/timing effect or initialization effect is plausible, but it was not separately tested, so no such mechanism is claimed as established.

Occlusion	Seed	Step	True normal	Estimated normal	Outcome
25%	2	6	(-0.957, -0.291)	(-0.988, -0.152)	Collision
25%	3	6	(-0.956, -0.294)	(-0.986, -0.164)	Collision
25%	4	6	(-0.956, -0.293)	(-0.974, -0.226)	Collision
50%	2	18	(-0.164, -0.987)	(0.164, -0.986)	No collision
50%	3	18	(0.026, -1.000)	(0.339, -0.941)	No collision
50%	4	18	(0.047, -0.999)	(0.283, -0.959)	No collision

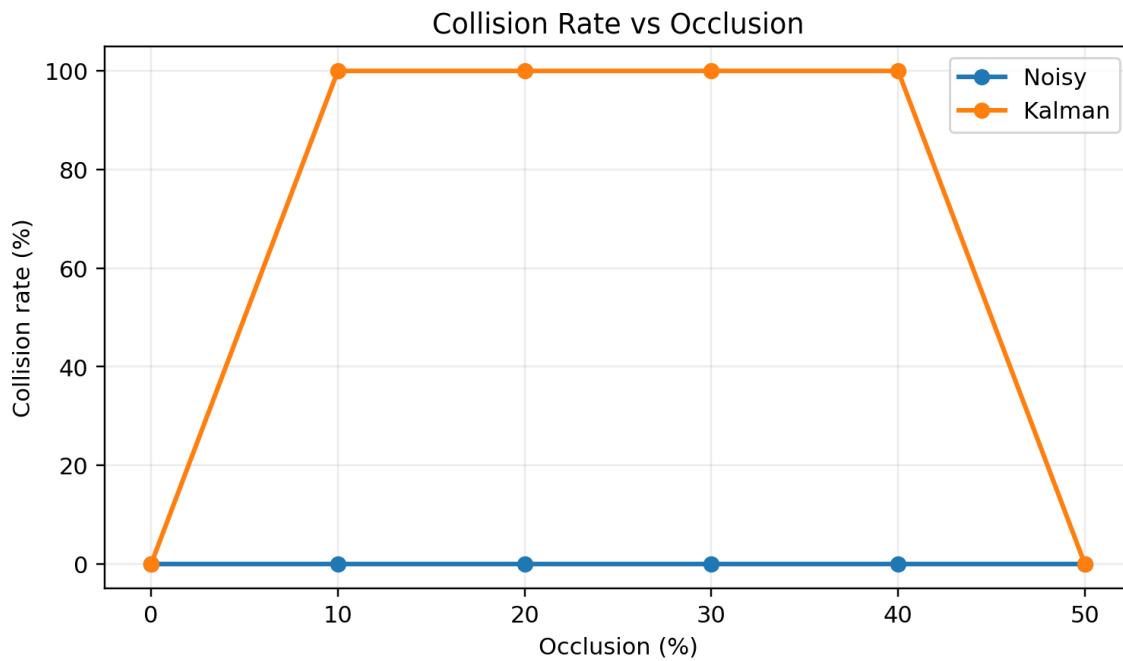


Table 1. True and estimated obstacle normals in representative occlusion cases (Kalman mode, 5 px pixel noise).

Figure 4. Collision rate vs occlusion.

Occlusion	Noisy RMSE	Kalman RMSE	Noisy clearance	Kalman clearance	Collision N	Collision K
0%	0.0000	0.0119	0.0933	0.0929	0%	0%
10%	0.4346	0.4350	0.2346	-0.0338	0%	100%
20%	0.5250	0.5253	0.2225	-0.0418	0%	100%
30%	0.6173	0.6176	0.2157	-0.0486	0%	100%
40%	0.7110	0.7113	0.2223	-0.0536	0%	100%
50%	0.3469	0.3471	0.2477	0.2537	0%	0%

5.6 Noise × Occlusion

The combined experiment showed that filtering benefit depends on the interaction between stochastic noise and systematic occlusion. At 5 px noise, RMSE reduction was 36.1% with no occlusion but only 1.6% at 25% occlusion and 4.6% at 50% occlusion. At 10 px noise the corresponding reductions were 36.1%,

5.4%, and 13.3%. Thus, the filter addressed stochastic noise effectively while leaving much of the systematic occlusion bias.

Noise	Occlusion	Noisy RMSE	Kalman RMSE	Reduction
0 px	0%	0.0000	0.0119	—
0 px	25%	0.5709	0.5712	-0.05%
0 px	50%	0.3469	0.3471	-0.06%
5 px	0%	0.1387	0.0887	36.1%
5 px	25%	0.5880	0.5784	1.6%
5 px	50%	0.3744	0.3571	4.6%
10 px	0%	0.2807	0.1794	36.1%
10 px	25%	0.6356	0.6013	5.4%
10 px	50%	0.4454	0.3861	13.3%

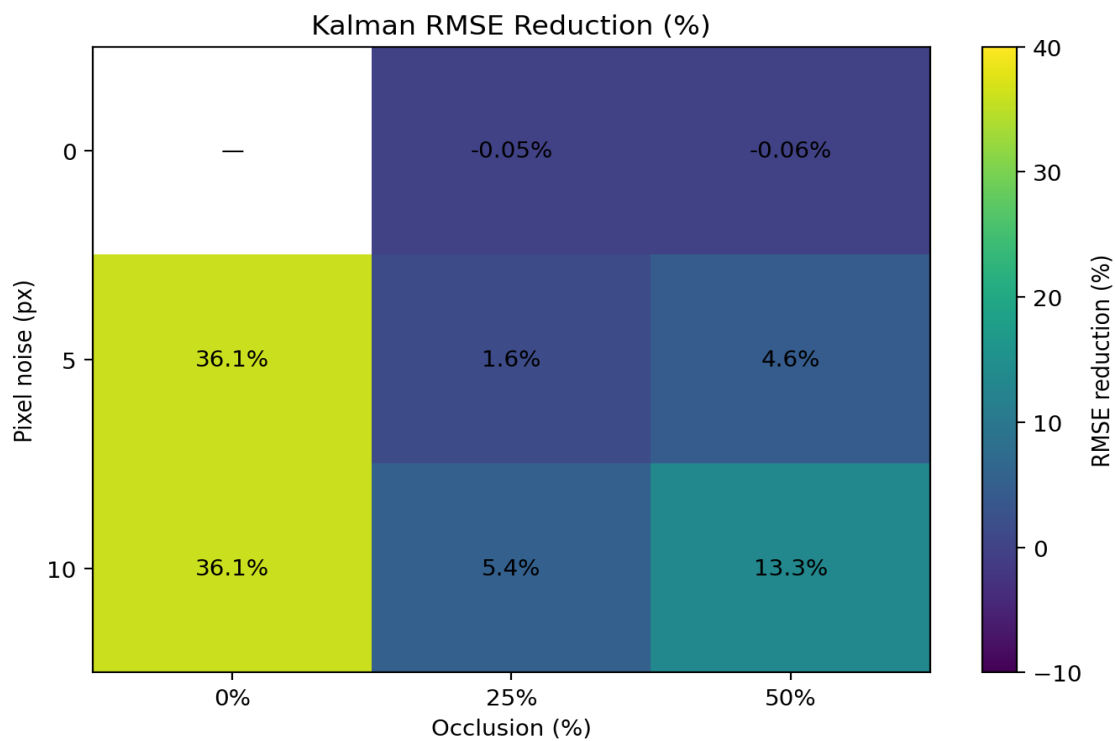


Figure 5. Kalman RMSE reduction across pixel-noise and occlusion levels.

6. Discussion

6.1 Perception-Control Interaction

The experiments demonstrate that perception quality is coupled to control performance. The estimated obstacle position directly affects the obstacle distance and local normal used by the DS modulation. Consequently, a perception error can alter both the magnitude and direction of the velocity modification.

6.2 Stochastic vs Systematic Error

Pixel noise is primarily stochastic localization error, making it suitable for temporal filtering. Occlusion is different: missing visual information can bias the detected centroid in a systematic direction. A Kalman filter designed to smooth noisy measurements cannot be expected to remove such bias reliably.

6.3 Temporal Resolution

The frame-rate results show that Kalman filtering is not independent of the perception update rate. At very low frame rates the prediction model operates for longer periods without measurement correction, and the filtered estimate can become less accurate than the raw measurement. In this static-obstacle setup, the degradation may possibly arise because the constant-velocity model extrapolates a noise-driven velocity estimate during these longer intervals. At 10–20 FPS, frequent updates allow the temporal model to provide substantial RMSE reduction.

6.4 Localization Accuracy vs Control Behavior

Reduced localization RMSE does not automatically imply improved convergence time, path length, or collision behavior. The controller reacts to the geometry and timing of the estimate, so the direction and temporal structure of an error can matter more than its aggregate magnitude.

7. ROS 2 / TurtleBot3 Demonstration

The research pipeline was subsequently integrated with ROS 2 and a TurtleBot3 Burger camera model in Gazebo. This stage was treated as system integration and demonstration rather than as a new statistical experiment.

Component	Implementation
Middleware	ROS 2 Jazzy
Simulator	Gazebo Sim 8.11.0
Robot	TurtleBot3 Burger camera model
Camera	/camera/image_raw
Odometry	/odom
Velocity	/cmd_vel
Perception	OpenCV + cv_bridge
State estimation	Kalman filter
Control	DS-modulation controller

The ROS implementation follows the same sequence: camera image → OpenCV obstacle detection → pixel-to-ground projection → Kalman filtering → world-frame obstacle estimate → DS-modulation control → velocity command. A single static obstacle and fixed goal were used for the demonstration. The obstacle was placed slightly off the direct robot-goal line to avoid the known directional ambiguity of the current DS implementation.

The final run successfully demonstrated obstacle detection, a change in robot trajectory around the obstacle, maintenance of separation during the avoidance motion, and subsequent goal reaching.

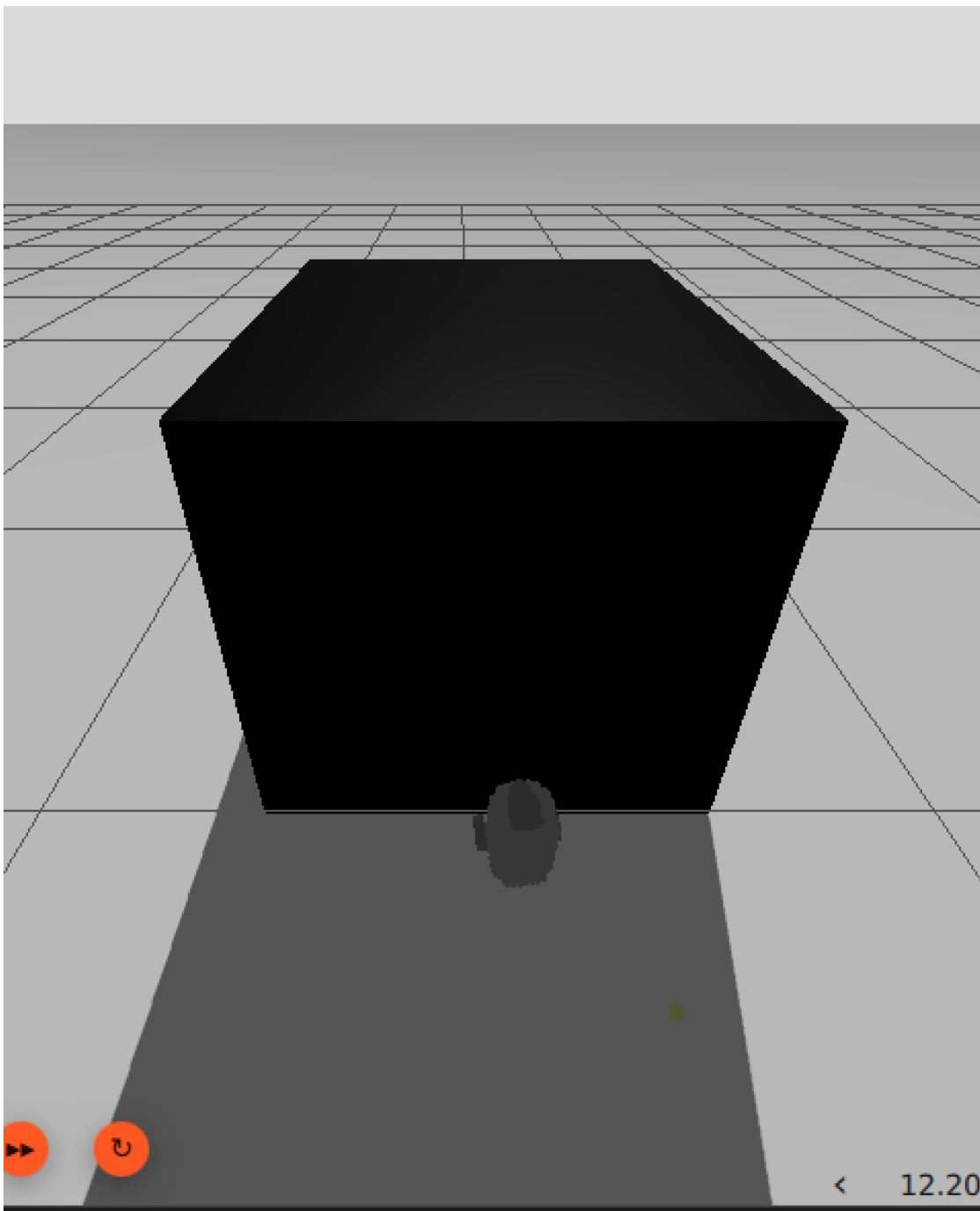


Figure 7.1. Gazebo Sim demonstration view showing the TurtleBot3 and the static obstacle used for the end-to-end ROS 2 run.

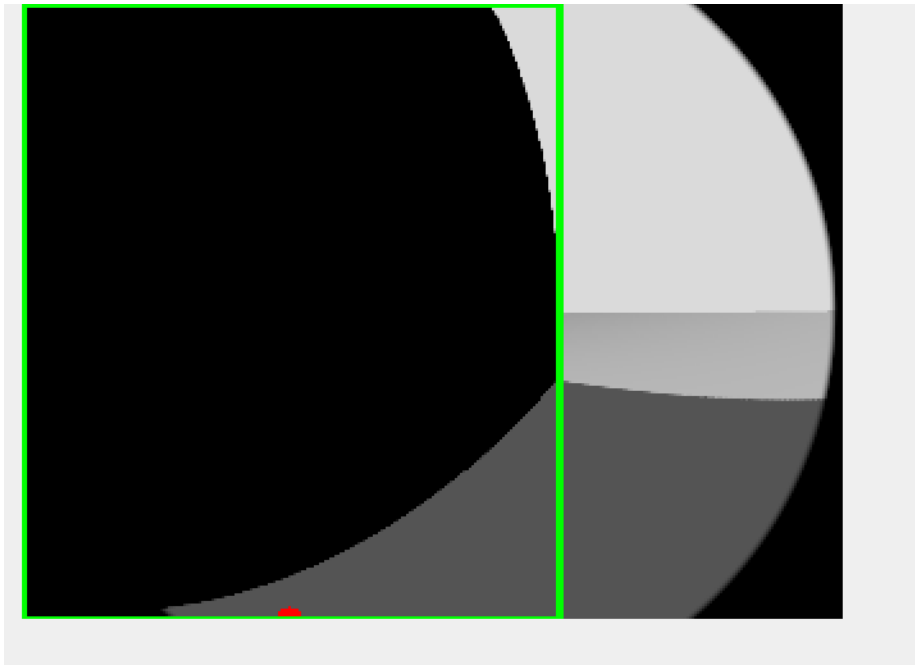


Figure 7.2. Camera-side detection output from the ROS 2 demonstration, with the detected obstacle region highlighted.

A separate demonstration video accompanies this report. It should be treated as supplementary material showing the working ROS 2/Gazebo system rather than as an additional statistical experiment.

8. Limitations

The latency sweep used a static obstacle, so localization RMSE is not expected to grow strongly with delay. Latency was therefore interpreted primarily through its effect on the robot's control response to delayed obstacle information.

- The main experiments use a synthetic camera and simulated robot rather than physical hardware.
- Perception degradation was introduced through controlled synthetic models and does not capture the full variability of real camera systems.

The Kalman filter uses a constant-velocity state model, but the experiments did not test a moving obstacle. The measured filtering benefit is therefore primarily evidence of noise averaging in a static-obstacle setting; the behavior of the chosen model under genuinely dynamic obstacles remains unvalidated.

- Several occlusion configurations produced deterministic repeated outcomes in the tested fixed geometry. Thus, the repeated 100-run summaries in the occlusion sweep demonstrate repeatability for the specified obstacle position and rectangular occlusion model rather than a robust estimate of collision probability across different geometries or environments.
- The controlled experiments use a simplified 2D single-obstacle environment and do not establish behavior in dense or dynamic scenes.
- The current DS implementation has a known directional ambiguity when robot, obstacle center, and goal are exactly collinear; those configurations were excluded.
- The ROS 2/Gazebo demonstration validates end-to-end integration but is not a statistical validation on physical hardware.

- The obstacle detector and camera projection are simplified and were designed for the controlled experimental environment.

9. Future Work

- Validate the pipeline with a physical mobile robot and real camera data.
- Replace synthetic degradation models with recorded, measured camera artifacts.
- Introduce uncertainty-aware obstacle representations so the controller can use perception confidence.
- Extend the system to multiple and dynamic obstacles.
- Investigate estimators or perception methods that explicitly address systematic occlusion bias.
- Study alternative state estimators and motion models under sparse/asynchronous perception.
- Combine perception uncertainty with formal safety constraints.

10. Conclusion

This project investigated how computer-vision perception quality affects DS-modulation-based obstacle avoidance. A complete pipeline was developed from image-based obstacle detection through geometric localization and Kalman filtering to reactive DS-based control. Controlled experiments showed that pixel noise increases localization error, low frame rates affect filtering effectiveness, latency reduces clearance and increases convergence time, and occlusion produces large systematic errors.

The Kalman filter consistently reduced localization RMSE under non-zero pixel noise and across the tested latency conditions. The latency result should not be interpreted as the filter explicitly compensating for delay: the obstacle was static, so delayed measurements did not change the underlying obstacle position error, while the robot-control response still degraded as delayed information was introduced. Improved localization accuracy did not always translate directly into better trajectory-level metrics. The frame-rate results showed that filtering effectiveness depends on temporal resolution, while the occlusion results demonstrated that smoothing alone cannot reliably compensate for systematic perception errors.

The ROS 2/Gazebo implementation further demonstrated that perception, estimation, and control can be connected into a real-time TurtleBot3 pipeline. Overall, the results support the central premise of the project: reactive obstacle avoidance depends not only on the controller, but also on the quality, timing, and structure of the perception information provided to it.

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- [11] ROBOTIS. TurtleBot3 documentation and simulation packages used for the Gazebo demonstration.

Appendix A. Experimental Parameter Summary

Parameter	Tested levels / setting	Fixed settings
Main sweep repetitions	100 runs per tested condition	—
Pixel noise	0, 2, 4, 6, 8, 10 px	20 FPS
Blur	0, 3, 5, 7, 9 px	5 px pixel noise
Frame rate	2, 4, 5, 10, 20 FPS	5 px pixel noise
Latency	0, 50, 100, 150, 200 ms	5 px pixel noise
Occlusion	0, 10, 20, 30, 40, 50%	0 px pixel noise
Noise × occlusion	0, 5, 10 px × 0, 25, 50%	Fewer seeds than main sweeps
Significance level	$\alpha = 0.05$	—
Multiple-comparison correction	Holm correction	Within each experiment
Estimator	Constant-velocity Kalman filter	—
Controller	DS-modulation with goal-directed nominal dynamics	—

Appendix B. Supplementary Demonstration Video

The video demonstrates the TurtleBot3 camera, obstacle detection, Kalman-filtered localization, DS-modulation avoidance, and goal reaching.